

1. **Summary:** Briefly summarize the paper and its contributions. This is not the place to critique the paper; the authors should generally agree with a well-written summary. This is also not the place to paste the abstract—please provide the summary in your own understanding after reading.

Point4D introduces a feed-forward architecture for long-range 4D motion reconstruction designed to overcome key bottlenecks in prior work: (1) trajectory loss under occlusion in 2D pixel-query methods, (2) memory limits when processing multi-hundred-frame videos in Transformer encoders, and (3) compounding reprojection errors during chunk-wise trajectory chaining. Given an input video, a shared ViT encoder extracts per-frame scene representations, camera poses, and depth maps. To decode motion, Point4D constructs 3D spatial coordinate queries $p = (x, y, z)$ at a source timestep, paired with a visual patch descriptor extracted once from a reference frame where the point is visible. The query embedding cross-attends to the chunk's scene representation to predict the point's 3D position at a target timestep. Because queries are defined in 3D physical space and visual descriptors are visibility-agnostic, tracking remains valid through occlusions and out-of-frame exits. To scale to long sequences, the model partitions video into overlapping temporal chunks, aligns chunk coordinate systems via least-squares Sim3 transformations, and directly re-queries predicted 3D endpoints across chunk boundaries without image-plane reprojection. The method was evaluated across three regimes: (1) long-video tracking (150–500 frames) to measure trajectory chaining, (2) single-chunk tracking (64 frames) to isolate raw motion decoder accuracy, and (3) video depth and camera pose estimation. Compared against feed-forward 4D models, iterative 3D trackers, and 2D+3D pipelines, Point4D achieved state-of-the-art performance on long sequences, exhibiting significantly slower accuracy degradation over sequential chunk handoffs and substantial gains under occlusion. Single-chunk benchmarks confirmed that Point4D's primary advantage stems specifically from its 3D query trajectory chaining framework rather than superior single-chunk decoder capacity.

2. **Strengths and Weaknesses:** Please provide a thorough assessment of the strengths and weaknesses of the paper. A good mental framing for strengths and weaknesses is to think of reasons you might accept or reject the paper. Please touch on the following dimensions:

1. *Quality:* Is the submission technically sound? Are claims well supported (e.g., by theoretical analysis or experimental results)? Are the methods used appropriate? Is this a complete piece of work or work in progress? Are the authors careful and honest about evaluating both the strengths and weaknesses of their work?

The submission is technically sound, rigorous, and represents a complete, well-validated work. The underlying architecture is mathematically grounded, combining a pretrained Vision Transformer backbone (*Depth Anything 3*), cross-attention 3D query decoding, multi-task supervision losses and Sim3 coordinate alignment. The empirical evaluation is also thorough, testing on multiple long range sequences from *PointOdyssey*, *Dynamic Replica*, and *Panoptic Studio* alongside geometry and pose benchmarks (*KITTI*, *Bonn*, *Sintel*). The targeted diagnostic ablations—such as single-chunk (64-frame) baseline comparisons and visibility regime breakdowns—also show that long-range performance gains stem specifically from the advancements in 3D query chaining. I think the methods used were appropriate; correspondingly, the choice of evaluation metrics—including 3D Endpoint Error (EPE), threshold accuracy (APD), and Survival Rate (SR)—effectively captures both fine-grained spatial accuracy and trajectory longevity across visible, occluded, and out-of-frame regimes. However I think there could have been some additional ablations for tuning chunk size and overlap sensitivity to measure how tracking performance, GPU memory usage, and inference time trade off across different chunk lengths. While standard depth and pose benchmarks verify the fidelity of the visual geometry backbone. Furthermore, evaluating tracking error against continuous occlusion span (e.g., EPE vs. occlusion duration in frames) and reporting standard 2D TAPVid metrics (such as Average Jaccard) via reprojection would provide a clearer picture of performance against established 2D tracking literature."

2. *Clarity*: Is the submission clearly written? Is it well organized? (If not, please make constructive suggestions for improving its clarity.) Does it adequately inform the reader? (Note that a superbly written paper provides enough information for an expert reader to reproduce its results.)

I think it is written clearly and is well organized.

3. *Significance*: Are the results impactful for the community? Are others (researchers or practitioners) likely to use the ideas or build on them? Does the submission address a difficult task in a better way than previous work? Does it advance our understanding/knowledge on the topic in a demonstrable way? Does it provide unique data, unique conclusions about existing data, or a unique theoretical or experimental approach?

The results presented in this paper seem to be impactful and advance the state of the art in long-range 4D motion reconstruction, the primary conceptual contribution—replacing traditional 2D pixel queries with 3D coordinate queries paired with a visibility-agnostic visual patch descriptor—fundamentally

decouples trajectory prediction from image-plane visibility. This key formulation addresses a major bottleneck in existing feed-forward 4D models (such as Any4D, 4RC, and V-DPM), which rely on 2D pixel reprojection across video chunks and subsequently fail during occlusions or field-of-view exits. This key shift also enables the autoregressive trajectory chaining in 3D physical space and coordinate alignment across overlapping 48-frame chunks method, which demonstrated positive results. Crucially, the experiments demonstrate that Point4D's tracking accuracy degrades significantly slower across sequential chunk handoffs than 2D-query baselines, while remaining computationally lighter and faster than memory-intensive iterative trackers like SpatialTrackerV2. Given this, researchers and practitioners in robotics, AR/VR, and 4D world modeling are very likely to build upon this approach more, especially because there is room to be improved with adding feature memory/storing information between chunks.

4. *Originality*: Does the work provide new insights, deepen understanding, or highlight important properties of existing methods? Is it clear how this work differs from previous contributions, with relevant citations provided? Does the work introduce novel tasks or methods that advance the field? Does this work offer a novel combination of existing techniques, and is the reasoning behind this combination well-articulated? **As the questions above indicates, originality does not necessarily require introducing an entirely new method. Rather, a work that provides novel insights by evaluating existing methods, or demonstrates improved efficiency, fairness, etc. is also equally valuable.**

Point4D demonstrates originality through a novel combination and reformulation of existing techniques. The backbone architecture extends existing work; the paper combines a feed-forward visual geometry encoder (initialized from Depth Anything 3) and cross-attention motion decoding (extending D4RT). The core technical novelty lies in replacing traditional 2D pixel queries with 3D spatial coordinate queries, paired with a visibility-agnostic visual patch descriptor. The ways in which the paper is built upon previous work and addresses the gaps that they have left is well articulated.

You can incorporate Markdown and LaTeX into your review. See <https://openreview.net/faq>.

3. **Quality**: Based on what you discussed in “Strengths and Weaknesses”, please assign the paper a numerical rating on the following scale to indicate the quality of the work.

1. 4 excellent

2. 3 good

3. 2 fair
4. 1 poor
4. **Clarity:** Based on what you discussed in “Strengths and Weaknesses”, please assign the paper a numerical rating on the following scale to indicate the clarity of the paper.
 1. 4 excellent
 2. 3 good
 3. 2 fair
 4. 1 poor
5. **Significance:** Based on what you discussed in “Strengths and Weaknesses”, please assign the paper a numerical rating on the following scale to indicate the significance of the paper.
 1. 4 excellent
 2. 3 good
 3. 2 fair
 4. 1 poor
6. **Originality:** Based on what you discussed in “Strengths and Weaknesses”, please assign the paper a numerical rating on the following scale to indicate the originality of the paper.
 1. 4 excellent
 2. 3 good
 3. 2 fair
 4. 1 poor
7. **Questions:** Please list up and carefully describe questions and suggestions for the authors, which should focus on key points (ideally around 3–5) that are actionable with clear guidance. Think of the things where a response from the author can change your opinion, clarify a confusion or address a limitation. You are strongly encouraged to state the clear criteria under which your evaluation score could increase or decrease. This can be very important for a productive rebuttal and discussion phase with the authors.
 - 1) The paper highlights that 3D queries decouple tracking from image-plane visibility, which is a major strength. However, the chunking mechanism relies on the current chunk's encoded representation. If an object remains hidden or out of frame for an entire 48-frame chunk, then the chunk contains zero visual features of that object. Without inter-chunk temporal memory or velocity/inertia priors, how does the model prevent the spatial trajectory from drifting during such full-chunk occlusions?
 - a) Please provide a breakdown plot or table showing Endpoint Error (EPE) as a function of **continuous occlusion duration**. (+score if the authors outline a clear mechanism to handle total-chunk occlusions or show the impact of longer occlusions on the metrics)
 - 2) Autoregressive trajectory chaining relies on least-squares Sim3coordinate alignment computed from predicted depth maps across overlapping boundary frames. On long sequences (e.g., 500 frames / 12 handoffs on PointOdyssey), how

can we know how much cumulative trajectory error stems from depth/pose misestimation versus actual motion decoder drift?

- a) Please include an ablation experiment comparing chunk chaining using predicted depth maps versus ground-truth depth maps for Sim3 alignment across overlapping frames (+score if the ablation confirms that the motion decoder maintains very low intrinsic drift, proving that depth estimation is the primary bottleneck).
- 3) The main experiments fix the chunk size to 48 frames with an 8-frame overlap but the sensitivity to these hyperparameter choices is not currently quantified.
 - a) Provide a sensitivity analysis over chunk sizes and overlap sizes, showing trade-offs between memory, speed, and chaining accuracy (+score if tracking performance remains robust across reasonable chunk parameters)
- 4) The paper focuses almost exclusively on 3D metrics (EPE, APD, Survival Rate) on sparse point sets. However, it does not evaluate standard 2D tracking metrics (such as Average Jaccard / AJ on TAPVid) via reprojection, nor does it quantify tracking accuracy when scaling to dense query sets.
 - a) Report projected 2D tracking metrics to enable direct comparisons with state-of-the-art pure 2D trackers (+score if projected 2D tracking accuracy remains competitive with top 2D trackers, and tracking precision remains stable when scaling to dense query point sets)

8. Limitations: Have the authors adequately addressed the limitations and potential negative societal impact of their work? If so, simply leave “yes”; if not, please include constructive suggestions for improvement. In general, authors should be rewarded rather than punished for being up front about the limitations of their work and any potential negative societal impact. You are encouraged to think through whether any critical points are missing and provide these as feedback for the authors.

While the authors acknowledge three main limitations—the lack of persistent feature memory across chunks, unreliability during full-chunk occlusions, and compounding depth estimation errors—their central claim regarding the effects of visibility decoupling warrants a more granular distinction. Specifically, Point4D’s 3D query framework dramatically improves trajectory prediction for short or transient occlusions where the target object appears elsewhere within the same 48-frame chunk. However, for prolonged occlusions exceeding the chunk window, the lack of inter-chunk motion memory (such as velocity or inertia priors) leaves the decoder without visual context or kinematic priors, causing spatial predictions to drift until the point re-emerges. I also think that an ethical discussion could be useful (but isn’t mentioned currently). It seems that the implications of this paper are that stepping out of frame or occluding oneself no longer reliably breaks camera tracking, which could raise concerns for long-range surveillance or tracking

without people's consent. Furthermore, high-precision 3D motion tracking has lots of applications in defense, particularly in autonomous targeting. Even more broadly, physical tracking inference without permission from monocular video could have unethical implications due to the high volume of existing consumer camera footage, where people may not consent to being tracked and profiled spatially. Therefore, I think that this paper could use a discussion on the potential for misuse and dual-use, which it currently doesn't have.

9. Overall: Please provide an "overall score" for this submission. Choices:

1. 6: Strong Accept: Technically flawless paper with groundbreaking impact on one or more areas of AI, with exceptionally strong evaluation, reproducibility, and resources, and no unaddressed ethical considerations.
2. 5: Accept: Technically solid paper, with high impact on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.
3. 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
4. 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.
5. 2: Reject: For instance, a paper with technical flaws, weak evaluation, inadequate reproducibility and incompletely addressed ethical considerations.
6. 1: Strong Reject: For instance, a paper with well-known results or unaddressed ethical considerations

10. Confidence: Please provide a "confidence score" for your assessment of this submission to indicate how confident you are in your evaluation. Choices

1. 5: You are absolutely certain about your assessment. You are very familiar with the related work and checked the math/other details carefully.
2. 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
3. 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
4. 2: You are willing to defend your assessment, but it is quite likely that you did not understand the central parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
5. 1: Your assessment is an educated guess. The submission is not in your area or the submission was difficult to understand. Math/other details were not carefully checked.

11. **Ethical concerns:** If there are ethical issues with this paper, please flag the paper for an ethics review. For guidance on when this is appropriate, please review the [NeurIPS ethics guidelines](#).
12. **Code of conduct acknowledgement.** While performing my duties as a reviewer (including writing reviews and participating in discussions), I have and will continue to abide by the NeurIPS code of conduct ([NeurIPS Code Of Conduct](#)).
13. **Responsible reviewing acknowledgement:** I acknowledge I have read the information about the "responsible reviewing initiatives" and will abide by that.
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